Část D: Limity funkcí a posloupností

1. Vyšetřování limit bez použití l'Hospitalova pravidla

D/ 1) Najděte limity:

a) $\lim\limits_{x\to\infty} \dfrac{2x^4-3x+5}{5x^2-6}$, b) $\lim\limits_{x\to\infty} \dfrac{4x^2+x+4}{x^3-6x}$, c) $\lim\limits_{x\to\infty} \dfrac{4x^2+x+4}{5x^2-6}$,

d) $\lim\limits_{x\to\infty} \dfrac{(2x^3+4x^2+5)(x^2+x+1)}{(x+2)(x^4+2x^3+7x^2+x-1)}$ (neroznásobujte závorky!) .

D/ 2) Najděte limity:

a) $\lim\limits_{x\to\infty} (\sqrt{2x^2+1}-\sqrt{x^2+1}\,)$, b) $\lim\limits_{x\to\infty} x(\sqrt[3]{x+1}-\sqrt[3]{x+2})$,

c) $\lim\limits_{n\to\infty} \dfrac{\sqrt{n^2+2}-2n}{\sqrt[3]{n^2+2-2n}}$, d) $\lim\limits_{n\to\infty} \dfrac{(n+2)!+2(n+1)!}{2(n+2)!-(n+1)!}$.

D/ 3) Najděte $\lim_{x\to x_0} f(x)$:

a) $f(x)=\dfrac{2x^2-11x+5}{3x^2-14x-5}$, $x_0=0;\ 5;\ -\infty$, b) $f(x)=\dfrac{x^3-x^2-x+1}{x^3-4x^2+5x-2}$, $x_0=1;\ 2^+;\ 2^-$,

c) $f(x)=\dfrac{3}{1-x^3}+\dfrac{1}{x-1}$, $x_0=0;\ 1;\ -\infty$.

D/ 4) Najděte $\lim_{x\to x_0} f(x)$:

a) $f(x)=\dfrac{1-x^2}{1-\sqrt{x}}$, $x_0=0^+;\ 0^-;\ 1;\ +\infty$, b) $f(x)=\dfrac{\sqrt{x^2-x-2}}{x+1}$, $x_0=+\infty;\ -\infty$,

c) $f(x)=\dfrac{1+\sqrt{2x^2-1}}{x}$, $x_0=0;\ +\infty;\ -\infty$, d) $f(x)=\dfrac{1}{\sqrt{x^2+8x+3}-\sqrt{x^2-8x+3}}$, $x_0=0;\ +\infty;\ -\infty$.

D/ 5) Určete $\lim_{x\to x_0} h(x)$ a o jaký typ limity se jedná:

a) $x_0=+\infty$, $h(x)=f(x)-g(x)$, kde $f(x)=x^2$ a

1) $g(x)=x$, 2) $g(x)=x^2+1/x$, 3) $g(x)=x^3$, 4) $g(x)=x^2-\cos(\pi x)$;

b) $x_0=+\infty$, $h(x)=f(x)\cdot g(x)$, kde $f(x)=x$ a

1) $g(x)=1/(\sqrt{x}+1)$, 2) $g(x)=1/(x+2)$, 3) $g(x)=1/(x^2-2)$,
4) $g(x)=-1/\sqrt{x}$, 5) $g(x)=(\cos 3x)/x$;

c) $x_0=+\infty$, $h(x)=f(x)/g(x)$, kde $f(x)=x^2$ a

1) $g(x)=x+5$, 2) $g(x)=x^2-x+1$, 3) $g(x)=x^3-2x^2$;

d) $x_0=+\infty$, $h(x)=f(x)/g(x)$, kde $f(x)=1/x^2$ a

1) $g(x)=1/(2x-1)$, 2) $g(x)=1/(x^2+8)$, 3) $g(x)=1/(3x^3-x+6)$,
4) $g(x)=-1/(3x^2-2)$, 5) $g(x)=-1/(x^3+x^2-8)$;

e) $x_0=0$, $h(x)=f(x)/g(x)$, kde $f(x)=2$ a

1) $g(x)=x^2/(1+2x^2)$, 2) $g(x)=x^2/(x-2)$, 3) $g(x)=x/(2x+3)$.

(Už víte, proč nedefinujeme $\infty-\infty$, $\infty\cdot 0$, ∞/∞, $0/0$, $a/0$?)

D/ 6) Najděte limity:

a) $\lim\limits_{x\to\infty} \dfrac{4^x+5^x}{4^{x+1}-5^{x+1}}$, b) $\lim\limits_{x\to\infty} \dfrac{4^{2x}+5^x}{4^{2x+1}-5^{x+1}}$, b) $\lim\limits_{n\to\infty} \dfrac{(-4)^{n-1}+7\cdot 3^{2n}}{5\cdot 3^{2n+2}+2\cdot(-7)^{n+2}}$.

D/ 7) Najděte limity:

a) $\lim\limits_{x\to\infty} \left(2+\dfrac{1}{x}\right)^x$, b) $\lim\limits_{x\to\infty} \left(1+\dfrac{4}{x}\right)^x$, c) $\lim\limits_{n\to\infty} \left(1-\dfrac{1}{n}\right)^n$,

d) $\lim\limits_{n\to\infty} \left(\dfrac{n^2+1}{n^2}\right)^{n^2+1}$, e) $\lim\limits_{n\to\infty} \left(\dfrac{2n-2}{2n-3}\right)^{6n-4}$ $(n\geq 2)$.

D/ 8) Najděte $\lim_{x\to x_0}\ f(x)$:

a) $f(x)=\dfrac{\cos 2x}{\sin x-\cos x}$, $x_0=0;\ \dfrac{\pi}{4}$, b) $f(x)=\operatorname{tg} x-\dfrac{1}{\cos x}$, $x_0=\dfrac{\pi}{2};\ \dfrac{\pi}{4}$,

c) $f(x)=\dfrac{\sin x}{\sqrt{1+\operatorname{tg} x}-\sqrt{1-\operatorname{tg} x}}$, $x_0=0;\ -\dfrac{\pi}{4}^+$.

D/ 9) Najděte $\lim_{x\to x_0}\ f(x)$:

a) $f(x)=\dfrac{x^2-1}{\sin(x+1)}$, $x_0=0;\ -1$, b) $f(x)=\dfrac{x^2}{\operatorname{tg}^2\left(\frac{x}{2}\right)}$, $x_0=0;\ \pi$,

c) $f(x)=\sin 3x\cdot\operatorname{cotg} 5x$, $x_0=0;\ \dfrac{\pi}{2}$.

D/ 10) Najděte $\lim_{x\to x_0}\ f(x)$:

a) $f(x)=(1+3\operatorname{tg} x)^{\operatorname{cotg} x}$, $x_0=\pi$, b) $f(x)=(1-2x)^{\frac{x+1}{x}}$, $x_0=0$.

D/ 11) Najděte limity funkce f v hraničních bodech množiny M:

a) $f(x)=\dfrac{\sqrt{x^2-9}-4}{x+5}$, $M=D(f)$, b) $f(x)=\dfrac{\sqrt{1+\operatorname{tg} x}-\sqrt{1+\sin x}}{x^3}$, $M=D(f)\cap\left\langle-\dfrac{\pi}{2},\dfrac{\pi}{2}\right\rangle$,

c) $f(x)=\left(\ln\dfrac{2x+1}{x}\right)^{\cos(\pi x)}$, $M=D(f)$.

D/ 12) Najděte limity:

a) $\lim\limits_{x\to 0}\ \operatorname{tg} x\cdot\operatorname{arctg}\dfrac{1}{x}$ b) $\lim\limits_{x\to 0}\left(x^2+\operatorname{arctg}\dfrac{1}{x}\right)$, c) $\lim\limits_{x\to 0}\left(\dfrac{1}{x^2}+\operatorname{arctg}\dfrac{1}{x}\right)$,

d) $\lim\limits_{x\to 0}\left(\operatorname{cotg} x+\operatorname{arctg}\dfrac{1}{x}\right)$, e) $\lim\limits_{x\to 0}\left(\operatorname{cotg} x\cdot\operatorname{arctg}\dfrac{1}{x}\right)$,

D/ 13) Najděte $\lim_{x\to x_0}\ f(x)$:

a) $f(x)=\cos x\cdot\operatorname{arccotg}\ x$, $x_0=+\infty;\ -\infty$, b) $f(x)=\dfrac{x^2+x-6}{x^2-5x+6}$, $x_0=2;\ 3$,

c) $f(x)=\dfrac{\sqrt{x^2+1}\ -x}{x}\cdot\cos\sqrt{x^2+1}$, $x_0=+\infty$, d) $f(x)=\dfrac{x^2+x-2}{\ln^2 x}$, $x_0=0^+;\ 1$.

D/ 14) Najděte limity:

a) $\lim\limits_{n\to\infty}\ \dfrac{n\cdot\operatorname{sgn}(-2)^n}{n^2+1}$, b) $\lim\limits_{n\to\infty}\ \dfrac{n\cdot\cos(n\pi)}{\sqrt{n^2+1}}$, c) $\lim\limits_{n\to\infty}\ (\ln 2)^n\cdot\sin(n^2)$,

d) $\lim\limits_{n\to\infty}\ n^2\cdot\operatorname{tg}(n\pi)$, e) $\lim\limits_{n\to\infty}\ \left(\cos(n^2\pi)+\dfrac{\pi}{n}\right)$, f) $\lim\limits_{n\to\infty}\ \left(\cos^2(n\pi)+\dfrac{\pi}{n}\right)$.

2. l'Hospitalovo pravidlo

D/ 15) Pomocí l'Hospitalova pravidla najděte $\lim_{x\to x_0} f(x)$:

a) $f(x) = \dfrac{1-\cos 2x}{1-\cos 5x}$, $x_0 = 0$,

b) $f(x) = \dfrac{x^4-16}{x^3+5x^2-6x-16}$, $x_0 = 2$,

c) $f(x) = x \cdot \operatorname{cotg} 2x$, $x_0 = 0$,

d) $f(x) = \left(\dfrac{1}{\sin x} - \dfrac{1}{x}\right)$, $x_0 = 0$,

e) $f(x) = x^{3/(x^2-1)}$, $x_0 = 1$,

f) $f(x) = \dfrac{3\cdot 4^x - 4\cdot 3^x}{5^x - 5\cdot 2^{x-1}}$, $x_0 = 1$.

D/ 16) Pomocí l'Hospitalova pravidla najděte $\lim_{x\to x_0} f(x)$:

a) $f(x) = \dfrac{\sqrt[3]{x}}{\ln^3 x}$, $x_0 = \infty$,

b) $f(x) = \dfrac{\sqrt[3]{\mathrm{e}^x}}{x^3}$, $x_0 = \infty$,

c) $f(x) = \sqrt[3]{x} \cdot \ln^3 x$, $x_0 = 0^+$,

d) $f(x) = \sqrt[3]{\mathrm{e}^x} \cdot x^3$, $x_0 = -\infty$.

D/ 17) Kde je to (rozumně) možné, spočítejte pomocí l'Hospitalova pravidla limity z části *1. Vyšetřování limit bez použití l'Hospitalova pravidla.*

(**Výsledky s návody jsou na dalších stránkách**.)

Výsledky:

D/1) a) $+\infty$ $\left(= \lim\limits_{x\to\infty} \dfrac{2x^2 - 3/x + 5/x^2}{5 - 6/x^2} = \left\langle\!\left\langle \dfrac{+\infty}{5} \right\rangle\!\right\rangle \right)$

b) 0 $\left(= \lim\limits_{x\to\infty} \dfrac{4/x + 1/x^2 + 4/x^3}{1 - 6/x^2} = \left\langle\!\left\langle \dfrac{0}{1} \right\rangle\!\right\rangle \right)$

c) $\dfrac{4}{5}$ $\left(= \lim\limits_{x\to\infty} \dfrac{4 + 1/x + 4/x^2}{5 - 6/x^2} \right)$

d) 2 $\Big($ nejvyšší mocnina x v čitateli i jmenovateli je pátá (vyjde pro $2x^3 \cdot x^2$ resp. $x \cdot x^4$), v čitateli je u ní koeficient 2, ve jmenovateli 1; stačí tedy zlomek vykrátit x^5 (tj. první závorku v čitateli vydělit x^3, druhou x^2 a první závorku ve jmenovateli vydělit x, druhou x^4) $\Big)$

D/2) a) $+\infty$ $\left(= \lim\limits_{x\to\infty} \dfrac{(\sqrt{2x^2+1} - \sqrt{x^2+1})(\sqrt{2x^2+1} + \sqrt{x^2+1})}{\sqrt{2x^2+1} + \sqrt{x^2+1}} = \lim\limits_{x\to\infty} \dfrac{(2x^2+1) - (x^2+1)}{(\sqrt{2x^2+1} + \sqrt{x^2+1})} = \right.$

$= \lim\limits_{x\to\infty} \dfrac{x^2}{|x|(\sqrt{2 + 1/x^2} + \sqrt{1 + 1/x^2})} = \lim\limits_{x\to\infty} \dfrac{x}{(\sqrt{2 + 1/x^2} + \sqrt{1 + 1/x^2})} = \left\langle\!\left\langle \dfrac{+\infty}{\sqrt{2}+1} \right\rangle\!\right\rangle \Big)$

b) $-\infty$ $\left(= \lim\limits_{x\to\infty} x \cdot \dfrac{(\sqrt[3]{x+1} - \sqrt[3]{x+2})(\sqrt[3]{(x+1)^2} + \sqrt[3]{(x+1)(x+2)} + \sqrt[3]{(x+2)^2}\ )}{\sqrt[3]{(x+1)^2} + \sqrt[3]{(x+1)(x+2)} + \sqrt[3]{(x+2)^2}} = \right.$

$= \lim\limits_{x\to\infty} x \cdot \dfrac{(x+1) - (x+2)}{\sqrt[3]{(x+1)^2} + \sqrt[3]{(x+1)(x+2)} + \sqrt[3]{(x+2)^2}} =$

$= \lim\limits_{x\to\infty} x \cdot \dfrac{-1}{x^{2/3} \cdot (\sqrt[3]{(1+1/x)^2} + \sqrt[3]{(1+1/x)(1+2/x)} + \sqrt[3]{(1+2/x)^2}\ )} =$

$= \lim\limits_{x\to\infty} x^{1/3} \cdot \dfrac{-1}{(\sqrt[3]{(1+1/x^3)^2} + \sqrt[3]{(1+1/x)(1+2/x)} + \sqrt[3]{(1+2/x)^2}\ )} = \left\langle\!\left\langle +\infty \cdot \dfrac{-1}{1+1+1} \right\rangle\!\right\rangle \Big)$

c) $-\infty$ $\left(= \lim\limits_{n\to\infty} \dfrac{n(\sqrt{1+2/n^2} - 2)}{x^{2/3}\sqrt[3]{1 + 2/x^2 - 2/x}} = \lim\limits_{n\to\infty} \left(n^{1/3} \cdot \dfrac{\sqrt{1+2/n^2} - 2}{\sqrt[3]{1 + 2/n^2 - 2/n}} \right) = \left\langle\!\left\langle +\infty \cdot \dfrac{-1}{1} \right\rangle\!\right\rangle \right)$

d) $\dfrac{1}{2}$ $\left(= \lim\limits_{n\to\infty} \dfrac{(n+1)!\ ((n+2)+2)}{(n+1)!\ (2(n+2)-1)} = \lim\limits_{n\to\infty} \dfrac{n+4}{2n+3} = \lim\limits_{n\to\infty} \dfrac{1+4/n}{2+3/n} \right)$

D/3) a)
- -1 pro $x_0 = 0$ $\Big(f(0) = -1\Big)$
- $\dfrac{9}{16}$ pro $x_0 = 5$ $\Big($typ $\left\langle\!\left\langle \dfrac{0}{0} \right\rangle\!\right\rangle$; $f(x) = \dfrac{(2x-1)(x-5)}{(3x+1)(x-5)}$ – viz zdůvodnění v b) pro $x_0 = 1$ $\Big)$
- $\dfrac{2}{3}$ pro $x_0 = -\infty$ $\Big($typ $\left\langle\!\left\langle \dfrac{\infty}{\infty} \right\rangle\!\right\rangle$; $f(x) = \dfrac{2 - 11/x + 5/x^2}{3 - 14/x - 5/x^2}\Big)$

b)
- -2 pro $x_0 = 1$ $\Big($typ $\left\langle\!\left\langle \dfrac{0}{0} \right\rangle\!\right\rangle$; protože 1 je kořenem čitatele i jmenovatele, máme v čitateli i jmenovateli kořenový činitel $(x-1)$ - vykrátíme jím – opět dostaneme limitu typu $\left\langle\!\left\langle \dfrac{0}{0} \right\rangle\!\right\rangle$ - postup opakujeme, pak dosadíme: $f(x) = \dfrac{(x-1)(x^2-1)}{(x-1)(x^2-3x+2)} = \dfrac{x^2-1}{x^2-3x+2} = \dfrac{(x-1)(x+1)}{(x-1)(x-2)} = \dfrac{x+1}{x-2}\Big)$
- $+\infty$ pro $x_0 = 2^+$, $-\infty$ pro $x_0 = 2^-$, $\Big($v obou případech typ $\left\langle\!\left\langle \dfrac{3}{0} \right\rangle\!\right\rangle$; jako výše $f(x) = \dfrac{x+1}{x-2}$, tedy pro $x_0 = 2^+$ máme limitu typu $\left\langle\!\left\langle \dfrac{3}{0^+} \right\rangle\!\right\rangle$, pro $x_0 = 2^-$ typu $\left\langle\!\left\langle \dfrac{3}{0^-} \right\rangle\!\right\rangle$ $\Big)$

c)
- 2 pro $x_0 = 0$ $\Big(f(0) = 2\Big)$
- 1 pro $x_0 = 1$ $\Big($typ $\left\langle\!\left\langle \dfrac{3}{0} + \dfrac{1}{0} \right\rangle\!\right\rangle$, upravíme: $f(x) = \dfrac{3 - (x^2+x+1)}{1-x^3}$ - typ $\left\langle\!\left\langle \dfrac{0}{0} \right\rangle\!\right\rangle$, jako v b) rozložíme čitatele a jmenovatele, vykrátíme - dostaneme $f(x) = \dfrac{x+2}{x^2+x+1}$ - a dosadíme $\Big)$
- 0 pro $x_0 = -\infty$ $\Big($typ $\left\langle\!\left\langle 0 + 0 \right\rangle\!\right\rangle$ $\Big)$

D/4) a)
- 1 pro $x_0 = 0^+$ $\Big(f(0) = 1\Big)$
- nemá smysl pro $x_0 = 0^-$ $\Big($ f není definována pro $x < 0$ $\Big)$

- 4 pro $x_0 = 1$ $\Big($typ $\Big\langle\!\Big\langle \frac{0}{0} \Big\rangle\!\Big\rangle$; $f(x) = \dfrac{(1-x)(1+x)}{1-\sqrt{x}} = \dfrac{(1-\sqrt{x})(1+\sqrt{x})(1+x)}{1-\sqrt{x}} = (1+\sqrt{x})(1+x)$ $\Big)$
- $+\infty$ pro $x_0 = +\infty$ $\Big(f(x) = \dfrac{x^2(1/x^2-1)}{\sqrt{x}(1/\sqrt{x}-1)} = \dfrac{x^{3/2}(1/x^2-1)}{(1/\sqrt{x}-1)} = \Big\langle\!\Big\langle \dfrac{\infty\cdot(0-1)}{0-1} \Big\rangle\!\Big\rangle$
  nebo f upravit jako pro $x_0 = 1$ $\Big)$

b)
- 1 pro $x_0 = +\infty$, -1 pro $x_0 = -\infty$
  $\Big(f(x) = \dfrac{\sqrt{x^2(1-1/x-2/x^2)}}{x+1} = \dfrac{|x|\sqrt{1-1/x-2/x^2}}{x(1+1/x)} = \operatorname{sgn} x \cdot \dfrac{\sqrt{1-1/x-2/x^2}}{(1+1/x)}$ $\Big)$

c)
- nemá smysl pro $x_0 = 0$ $\Big($ f není definována na okolí nuly $\Big)$
- $\sqrt{2}$ pro $x_0 = +\infty$, $-\sqrt{2}$ pro $x_0 = -\infty$,
  $\Big(f(x) = \dfrac{1+\sqrt{x^2}\sqrt{2-1/x^2}}{x} = \dfrac{1}{x} + \dfrac{|x|\sqrt{2-1/x^2}}{x} = \dfrac{1}{x} + \operatorname{sgn} x \cdot \sqrt{2-1/x^2}\Big)$

d)
- neexistuje pro $x_0 = 0$; $+\infty$ pro $x_0 = 0^+$, $-\infty$ pro $x_0 = 0^-$,
  $\Big(f(x) = \dfrac{\sqrt{x^2+8x+3}+\sqrt{x^2-8x+3}}{(x^2+8x+3)-(x^2-8x+3)} = \dfrac{\sqrt{x^2+8x+3}+\sqrt{x^2-8x+3}}{16x} = \Big\langle\!\Big\langle \dfrac{2\sqrt{3}}{16\cdot 0^{\pm}} \Big\rangle\!\Big\rangle$ $\Big)$
- $\dfrac{1}{8}$ pro $x_0 = +\infty$, $-\dfrac{1}{8}$ pro $x_0 = -\infty$,
  $\Big(f(x) = \dfrac{\sqrt{x^2+8x+3}+\sqrt{x^2-8x+3}}{(x^2+8x+3)-(x^2-8x+3)} = \dfrac{|x|(\sqrt{1+8/x+3/x^2}+\sqrt{1-8/x+3/x^2})}{16x} =$
  $= \operatorname{sgn} x \cdot \dfrac{(\sqrt{1+8/x+3/x^2}+\sqrt{1-8/x+3/x^2})}{16}$ $\Big)$

D/5) a) 1) ∞, 2) 0, 3) $-\infty$, 4) neexistuje (všechno jsou limity typu $\Big\langle\!\Big\langle \infty - \infty \Big\rangle\!\Big\rangle$)

b) 1) ∞, 2) 1, 3) 0, 4) $-\infty$, 5) neexistuje (všechno jsou limity typu $\Big\langle\!\Big\langle \infty \cdot 0 \Big\rangle\!\Big\rangle$)

c) 1) ∞, 2) 1, 3) 0 (všechno jsou limity typu $\Big\langle\!\Big\langle \infty/\infty \Big\rangle\!\Big\rangle$)

d) 1) 0, 2) 1, 3) ∞, 4) -3, 5) $-\infty$ (všechno jsou limity typu $\Big\langle\!\Big\langle 0/0 \Big\rangle\!\Big\rangle$)

e) 1) ∞, 2) $-\infty$, 3) neexistuje (všechno jsou limity typu $\Big\langle\!\Big\langle 2/0 \Big\rangle\!\Big\rangle$)

D/6) a) $-\dfrac{1}{5}$ $\Big(f(x) = \dfrac{5^x((4/5)^x+1)}{5^x(4(4/5)^x-5)} = \dfrac{(4/5)^x+1}{4(4/5)^x-5}$; limita typu: $\Big\langle\!\Big\langle \dfrac{0+1}{4\cdot 0-5} \Big\rangle\!\Big\rangle\Big)$

b) $\dfrac{1}{4}$ $\Big(f(x) = \dfrac{16^x+5^x}{4\cdot 16^x - 5\cdot 5^x} = \dfrac{1+(5/16)^x}{4-5(5/16)^x}$; limita typu: $\Big\langle\!\Big\langle \dfrac{1+0}{4-5\cdot 0} \Big\rangle\!\Big\rangle\Big)$

c) $\dfrac{7}{45}$ $\Big(a_n = \dfrac{(-4)^{n-1}+7\cdot 9^n}{5\cdot 9^{n+1}+2\cdot(-7)^{n+2}} = \dfrac{(-4)^{-1}(-4/9)^n+7}{5\cdot 9+2\cdot 7^2(-7/9)^n}$; limita typu: $\Big\langle\!\Big\langle \dfrac{(-4)^{-1}\cdot 0+7}{45+2\cdot 7^2\cdot 0} \Big\rangle\!\Big\rangle\Big)$

D/7) a) ∞ $\Big($ $f(x) > 2^x$ a $2^x \to \infty\Big)$

b) e^4 $\Big(f(x) = \Big(\Big(1+\dfrac{4}{x}\Big)^{x/4}\Big)^4 = \Big(\Big(1+\dfrac{1}{y}\Big)^y\Big)^4$, kde: $y = y(x) = \dfrac{x}{4} \longrightarrow \infty$ pro $x \to \infty$, a využijeme toho, že $\lim\limits_{y\to\infty}\Big(1+\dfrac{1}{y}\Big)^y = \mathrm{e}$ podle přehledu užitečných limit na konci Kapitoly 3 přednášek$\Big)$

c) e^{-1} $\left(\text{pro } n>1:\ a_n = \Big(\dfrac{n-1}{n}\Big)^n = \dfrac{1}{\Big(\frac{n}{n-1}\Big)^n} = \dfrac{1}{\Big(1+\frac{1}{n-1}\right)^{n-1}\Big(1+\frac{1}{n-1}\Big)}$; limita typu: $\Big\langle\!\Big\langle \dfrac{1}{\mathrm{e}\cdot 1} \Big\rangle\!\Big\rangle\right)$

d) e $\left(a_n = \Big(1+\dfrac{1}{n^2}\Big)^{n^2}\cdot\Big(1+\dfrac{1}{n^2}\Big)$; limita typu: $\Big\langle\!\Big\langle$vybraná posloupnost z $(1+1/k)^k\cdot 1\Big\rangle\!\Big\rangle\right)$

e) e^3 $\left(a_n = \Big(1+\dfrac{1}{2n-3}\Big)^{6n-4} = \left(\Big(1+\dfrac{1}{2n-3}\Big)^{(2n-3)}\right)^3\cdot\Big(1+\dfrac{1}{2n-3}\Big)^5$;

limita typu: $\Big\langle\!\Big\langle\Big($vybraná posloupnost z $(1+1/k)^k\Big)^3\cdot 1^5\Big\rangle\!\Big\rangle\right)$

D/8) a) • -1 pro $x_0 = 0$ $\Big(f(0) = -1\Big)$

• $-\sqrt{2}$ pro $x_0 = \dfrac{\pi}{4}$ $\Big($typ $\Big\langle\!\Big\langle \dfrac{0}{0} \Big\rangle\!\Big\rangle$; $f(x) = \dfrac{\cos^2 x - \sin^2 x}{\sin x - \cos x} = -(\cos x + \sin x)$ $\Big)$

b) • 0 pro $x_0 = \dfrac{\pi}{2}$ $\Big($typ $\Big\langle\!\Big\langle$neex. $-$ neex.$\Big\rangle\!\Big\rangle$; $f(x) = \dfrac{\sin x}{\cos x} - \dfrac{1}{\cos x} = \dfrac{\sin x - 1}{\cos x} = \dfrac{\sin^2 x - 1}{\cos x(\sin x + 1)} = \dfrac{-\cos x}{\sin x + 1}\Big)$

• $1 - \sqrt{2}$ pro $x_0 = \dfrac{\pi}{4}$ $\Big(f\Big(\dfrac{\pi}{4}\Big) = 1 - \dfrac{1}{\sqrt{2}/2}\Big)$

c) • 1 pro $x_0 = 0$ $\Big($typ $\Big\langle\!\Big\langle \dfrac{0}{0} \Big\rangle\!\Big\rangle$; $f(x) = \dfrac{\sin x(\sqrt{1+\operatorname{tg} x} + \sqrt{1-\operatorname{tg} x})}{(1+\operatorname{tg} x) - (1-\operatorname{tg} x)} = \dfrac{\sin x(\sqrt{1+\operatorname{tg} x} + \sqrt{1-\operatorname{tg} x})}{2\operatorname{tg} x} =$
$= \dfrac{\sin x(\sqrt{1+\operatorname{tg} x} + \sqrt{1-\operatorname{tg} x})}{2\sin x/\cos x} = \dfrac{(\sqrt{1+\operatorname{tg} x} + \sqrt{1-\operatorname{tg} x})\cos x}{2}$ $\Big)$

• $\dfrac{1}{2}$ pro $x_0 = -\dfrac{\pi}{4}{}^{+}$ $\Big($typ $\Big\langle\!\Big\langle \dfrac{-\sqrt{2}/2}{\sqrt{1+(-1)^+} - \sqrt{1-(-1)^+}} \Big\rangle\!\Big\rangle = \Big\langle\!\Big\langle \dfrac{-\sqrt{2}/2}{\sqrt{0^+} - \sqrt{2}} \Big\rangle\!\Big\rangle$ $\Big)$

D/9) a) • $-\dfrac{1}{\sin 1}$ pro $x_0 = 0$ $\Big(f(0) = -\dfrac{1}{\sin 1}\Big)$

• -2 pro $x_0 = -1$ $\Big($typ $\Big\langle\!\Big\langle \dfrac{0}{0} \Big\rangle\!\Big\rangle$; $f(x) = (x-1)\dfrac{x+1}{\sin(x+1)} = (x-1)\dfrac{1}{\sin(x+1)/(x+1)} = \Big\langle\!\Big\langle \dfrac{-2}{1} \Big\rangle\!\Big\rangle$,
protože $(x+1) \to 0$ pro $x \to -1$ a $\dfrac{\sin y}{y} \to 1$ pro $y \to 0$ $\Big)$

b) • 4 pro $x_0 = 0$ $\Big($typ $\Big\langle\!\Big\langle \dfrac{0}{0} \Big\rangle\!\Big\rangle$; $f(x) = \dfrac{x^2}{\big(\sin^2(\frac{x}{2})\big)/\big(\cos^2(\frac{x}{2})\big)} = \dfrac{(\frac{x}{2})^2 \cdot 4 \cdot \cos^2(\frac{x}{2})}{\sin^2(\frac{x}{2})} =$
$= \Big(\dfrac{\frac{x}{2}}{\sin(\frac{x}{2})}\Big)^2 \cdot 4 \cdot \cos^2\big(\dfrac{x}{2}\big) \to 1^2 \cdot 4 \cdot 1^2$ $\Big)$

• 0 pro $x_0 = \pi$ $\Big($typ $\Big\langle\!\Big\langle \dfrac{\pi^2}{(\pm\infty)^2} \Big\rangle\!\Big\rangle = \Big\langle\!\Big\langle \dfrac{\pi^2}{\infty} \Big\rangle\!\Big\rangle$ $\Big)$

c) • $\dfrac{3}{5}$ pro $x_0 = 0$ $\Big($typ $\Big\langle\!\Big\langle 0 \cdot (\pm\infty) \Big\rangle\!\Big\rangle$; $f(x) = \sin 3x \cdot \dfrac{\cos 5x}{\sin 5x} = \Big(\dfrac{\sin 3x}{3x} \cdot 3\Big)\Big(\dfrac{5x}{\sin 5x} \cdot \dfrac{1}{5}\Big) \cdot \cos 5x \to$
$\to 1 \cdot 3 \cdot 1 \cdot \dfrac{1}{5} \cdot 1\Big)$

• 0 pro $x_0 = \dfrac{\pi}{2}$ $\Big(f\Big(\dfrac{\pi}{2}\Big) = -1 \cdot 0\Big)$

D/10) a) e^3 $\Big(f(x) = \exp(\operatorname{cotg} x \cdot \ln(1 + 3\operatorname{tg} x)) = \exp(h(x)) = \mathrm{e}^{h(x)}$, přitom $\lim\limits_{x \to \pi^{\pm}} h(x) = \Big\langle\!\Big\langle (\pm\infty) \cdot 0 \Big\rangle\!\Big\rangle =$
$= \lim\limits_{x \to \pi^{\pm}} \dfrac{\ln(1+3\operatorname{tg} x)}{\operatorname{tg} x} = \lim\limits_{x \to \pi^{\pm}} \dfrac{\ln(1+3\operatorname{tg} x)}{3\operatorname{tg} x} \cdot 3 = 3$, protože $3\operatorname{tg} x \to 0$ pro $x \to \pi$ a $\dfrac{\ln(1+t)}{t} \to 1$
pro $t \to 0$ $\Big)$

b) e^{-2} $\Big(f(x) = \exp\Big(\Big(\dfrac{x+1}{x}\Big) \cdot \ln(1-2x)\Big) = \mathrm{e}^{h(x)}$, přitom $\lim\limits_{x \to 0} h(x) = \Big\langle\!\Big\langle \dfrac{1}{0} \cdot 0 \Big\rangle\!\Big\rangle =$
$= \lim\limits_{x \to 0} \Big(\dfrac{\ln(1-2x)}{(1-2x)-1} \cdot ((1-2x)-1) \cdot \dfrac{x+1}{x}\Big) = \lim\limits_{x \to 0} \Big(\dfrac{\ln(1-2x)}{(1-2x)-1} \cdot (-2) \cdot (x+1)\Big) = 1 \cdot (-2) \cdot 1,$
protože $(1-2x) \to 1$ pro $x \to 0$ a $\dfrac{\ln t}{t-1} \to 1$ pro $t \to 1$ $\Big)$

D/11) a) $M = (-\infty; -5) \cup (-5; -3\rangle \cup \langle 3; \infty)$ $\Big(= \big((-\infty; -3\rangle \cup \langle 3; \infty)\big) \setminus \{-5\}$ $\Big)$

• -1 pro $x_0 = -\infty$ $\Big($typ $\Big\langle\!\Big\langle \dfrac{\infty}{-\infty} \Big\rangle\!\Big\rangle$; $f(x) = \dfrac{|x|(\sqrt{1 - 9/x^2} - 4/|x|)}{x(1 + 5/x)} = \operatorname{sgn} x \cdot \dfrac{\sqrt{1 - 9/x^2} - 4/|x|}{1 + 5/x}$ $\Big)$

• $-\dfrac{5}{4}$ pro $x_0 = -5$ $\Big($typ $\Big\langle\!\Big\langle \dfrac{0}{0} \Big\rangle\!\Big\rangle$; $f(x) = \dfrac{(\sqrt{x^2-9} - 4)(\sqrt{x^2-9} + 4)}{(x+5)(\sqrt{x^2-9} + 4)} = \dfrac{(x^2-9) - 16}{(x+5)(\sqrt{x^2-9} + 4)} =$
$= \dfrac{x^2 - 25}{(x+5)(\sqrt{x^2-9} + 4)} = \dfrac{x-5}{\sqrt{x^2-9} + 4}$ $\Big)$

• -2 pro $x_0 = -3^-$ $\Big(f(-3) = -2\Big)$

• $-\dfrac{1}{2}$ pro $x_0 = 3^+$ $\Big(f(3) = -\dfrac{1}{2}\Big)$

• 1 pro $x_0 = \infty$ $\Big($ jako pro $x_0 = -\infty$, není tu ale nutné psát absolutní hodnotu $\Big)$

b) $M = \langle -\frac{\pi}{4}; 0) \cup (0; \frac{\pi}{2}\,)$ $\Big($ $= \langle -\frac{\pi}{4}; \frac{\pi}{2}\,) \setminus \{0\}$ (musí byt $\operatorname{tg} x \geq -1$) $\Big)$

- $\dfrac{\sqrt{2-\sqrt{2}}}{\sqrt{2}\,(\frac{\pi}{4})^3}$ pro $x_0 = -\dfrac{\pi}{4}^+$ $\Big(f\Big(-\dfrac{\pi}{4}\Big) = \dfrac{0-\sqrt{1-\sqrt{2}/2}}{(-\frac{\pi}{4})^3}$ $\Big)$
- $\dfrac{1}{4}$ pro $x_0 = 0$ $\Big($typ $\Big\langle\!\Big\langle \dfrac{1-1}{0} \Big\rangle\!\Big\rangle$; $f(x) = \dfrac{(1+\operatorname{tg} x)-(1+\sin x)}{x^3(\sqrt{1+\operatorname{tg} x}+\sqrt{1+\sin x})} =$

$= \dfrac{\operatorname{tg} x - \sin x}{x^3(\sqrt{1+\operatorname{tg} x}+\sqrt{1+\sin x})} = \dfrac{\sin x(1-\cos x)}{x^3\cos x(\sqrt{1+\operatorname{tg} x}+\sqrt{1+\sin x})} =$

$= \dfrac{\sin x(1-\cos^2 x)}{x^3\cos x(1+\cos x)(\sqrt{1+\operatorname{tg} x}+\sqrt{1+\sin x})} = \Big(\dfrac{\sin x}{x}\Big)^3 \dfrac{1}{\cos x(1+\cos x)(\sqrt{1+\operatorname{tg} x}+\sqrt{1+\sin x})}$ $\Big)$

- $+\infty$ pro $x_0 = \dfrac{\pi}{2}^-$ $\Big($typ $\Big\langle\!\Big\langle \dfrac{\sqrt{1+\infty}-\sqrt{2}}{(\pi/2)^3} \Big\rangle\!\Big\rangle$ $\Big)$

c) $M = (-\infty; -1) \cup (0; \infty)$ $\Big(f(x) = \exp\big(\cos \pi x \ \ln(\ln \frac{2x+1}{x}\,)\big) = \exp(h(x))$, tedy musí být $\ln \frac{2x+1}{x} > 0$, tj. $\frac{2x+1}{x} > 1$, tj. $\frac{1}{x} > -1$ $\Big)$

- neexistuje pro $x_0 = \pm\infty$ $\Big($typ pro h: $\Big\langle\!\Big\langle$neex. $\cdot \ln(\ln 2)\Big\rangle\!\Big\rangle$ $\Big)$
- $+\infty$ pro $x_0 = -1^-$ $\Big($typ pro h: $\Big\langle\!\Big\langle(-1)\cdot\ln(\ln 1^+)\Big\rangle\!\Big\rangle = \Big\langle\!\Big\langle -1\cdot(-\infty)\Big\rangle\!\Big\rangle$; pro f: $\Big\langle\!\Big\langle \mathrm{e}^{+\infty}\Big\rangle\!\Big\rangle$ $\Big)$
- $+\infty$ pro $x_0 = 0^+$ $\Big($typ pro h: $\Big\langle\!\Big\langle 1\cdot\ln(\ln(2+\frac{1}{0^+}))\Big\rangle\!\Big\rangle = \Big\langle\!\Big\langle 1\cdot\ln(\ln\infty)\Big\rangle\!\Big\rangle = \Big\langle\!\Big\langle 1\cdot\infty\Big\rangle\!\Big\rangle$; pro f: $\Big\langle\!\Big\langle \mathrm{e}^{+\infty}\Big\rangle\!\Big\rangle\Big)$

D/ 12) a) 0 $\Big($ limita typu: $\Big\langle\!\Big\langle 0\cdot$omez.$\Big\rangle\!\Big\rangle\Big)$

b) neexistuje $\Big($ limita typu: $\Big\langle\!\Big\langle 0+$neex.$\Big\rangle\!\Big\rangle$, protože $\lim\limits_{x\to 0^\pm} \operatorname{arctg}\dfrac{1}{x} = \lim\limits_{y\to\pm\infty} \operatorname{arctg} y = \pm\dfrac{\pi}{2}$ $\Big)$

c) $+\infty$ $\Big($ podle Věty 3-12,b), protože $\dfrac{1}{x^2} + \operatorname{arctg}\dfrac{1}{x} \geq \dfrac{1}{x^2} - \dfrac{\pi}{2}$ a $\lim\limits_{x\to 0}\Big(\dfrac{1}{x^2} - \dfrac{\pi}{2}\Big) = +\infty$

NEBO: $\lim\limits_{x\to 0^\pm}\Big(\dfrac{1}{x^2} + \operatorname{arctg}\dfrac{1}{x}\Big) = +\infty \pm \dfrac{\pi}{2} = +\infty$, tedy $\lim\limits_{x\to 0}\Big(\dfrac{1}{x^2} + \operatorname{arctg}\dfrac{1}{x}\Big) = +\infty$ $\Big)$

d) neexistuje $\Big(\lim\limits_{x\to 0^\pm}\Big(\operatorname{cotg} x + \operatorname{arctg}\dfrac{1}{x}\Big) = \pm\infty \pm \dfrac{\pi}{2} = \pm\infty$ $\Big)$

e) $+\infty$ $\Big(\lim\limits_{x\to 0^\pm}\Big(\operatorname{cotg} x \cdot \operatorname{arctg}\dfrac{1}{x}\Big) = \pm\infty\cdot(\pm\dfrac{\pi}{2}\,) = +\infty$ $\Big)$

D/ 13) a)

- 0 pro $x_0 = +\infty$ $\Big($typ $\Big\langle\!\Big\langle$omez. $\cdot\, 0\Big\rangle\!\Big\rangle$ $\Big)$
- neexistuje pro $x_0 = -\infty$ $\Big($typ $\Big\langle\!\Big\langle$neex. $\cdot\, \pi\Big\rangle\!\Big\rangle$ $\Big)$

b) $f(x) = \dfrac{(x-2)(x+3)}{(x-2)(x-3)} = \dfrac{x+3}{x-3}$ pro $x \notin \{2; 3\}$

- -5 pro $x_0 = 2$
- neexistuje pro $x_0 = 3$ $\Big(\lim\limits_{x\to 3^-} f(x) = \Big\langle\!\Big\langle \dfrac{6}{0^-} \Big\rangle\!\Big\rangle = -\infty$, $\lim\limits_{x\to 3^+} f(x) = \Big\langle\!\Big\langle \dfrac{6}{0^+} \Big\rangle\!\Big\rangle = +\infty\Big)$

c)

- 0 $\Big($typ $\Big\langle\!\Big\langle 0\cdot$omez.$\Big\rangle\!\Big\rangle$; $\dfrac{\sqrt{x^2+1}-x}{x} = \dfrac{\sqrt{x^2+1}}{x} - 1 = \dfrac{x\sqrt{1+1/x^2}}{x} - 1 = \sqrt{1+1/x^2} - 1\Big)$

d)

- 0 pro $x_0 = 0^+$ $\Big($typ $\Big\langle\!\Big\langle \dfrac{-2}{+\infty} \Big\rangle\!\Big\rangle$ $\Big)$
- neexistuje pro $x_0 = 1$ $\Big($ podle Př. 3-5 v přednáškách je $\lim\limits_{x\to 1^\pm} f(x) = \Big\langle\!\Big\langle \dfrac{0}{0} \Big\rangle\!\Big\rangle = \lim\limits_{x\to 1^\pm} \dfrac{x-1}{\ln x}\cdot\dfrac{x+2}{\ln x} =$ $\Big\langle\!\Big\langle 1\cdot\dfrac{3}{0^\pm} \Big\rangle\!\Big\rangle = \pm\infty$, $\Big)$

D/ 14) a) 0 $\Big(a_n = \dfrac{n}{n^2+1}\cdot\operatorname{sgn}(-2)^n$; limita typu : $\Big\langle\!\Big\langle 0\cdot$ omez. $\Big\rangle\!\Big\rangle\Big)$

b) neex. $\Big(a_n = \dfrac{n}{\sqrt{n^2+1}}\cdot\cos n\pi$; limita typu : $\Big\langle\!\Big\langle 1\cdot$ neex. $\Big\rangle\!\Big\rangle\Big)$

c) 0 $\Big($ limita typu : $\Big\langle\!\Big\langle 0\cdot$ omez. $\Big\rangle\!\Big\rangle$, protože $|\log 2| < 1$ $\Big)$

d) 0 $\Big($jde o konstantní nulovou posloupnost, nemá tedy smysl limitu zkoumat jako typ $\Big\langle\!\Big\langle \infty\cdot 0\Big\rangle\!\Big\rangle\Big)$

e) neex. $\Big($ limita typu : $\Big\langle\!\Big\langle$ neex. $+\,0\Big\rangle\!\Big\rangle$, protože $\cos n^2\pi = (-1)^n$ (proč tato rovnost platí?)$\Big)$

f) 1 $\Big(a_n = \big((-1)^n\big)^2 + \dfrac{\pi}{n} = 1 + \dfrac{\pi}{n}\Big)$

D/15) a) $\frac{4}{25}$ $(2\times$ l'H$)$ $\Big(\lim\limits_{x\to 0} f(x) = \Big\langle\Big\langle \frac{0}{0} \Big\rangle\Big\rangle \overset{\text{l'H}}{=} \lim\limits_{x\to 0} \frac{2\sin 2x}{5\sin 5x} = \Big\langle\Big\langle \frac{0}{0} \Big\rangle\Big\rangle \overset{\text{l'H}}{=} \lim\limits_{x\to 0} \frac{4\cos 2x}{25\cos 5x}\Big)$

b) $\frac{16}{13}$ $(1\times$ l'H$)$ $\Big(\lim\limits_{x\to 2} f(x) = \Big\langle\Big\langle \frac{0}{0} \Big\rangle\Big\rangle \overset{\text{l'H}}{=} \lim\limits_{x\to 2} \frac{4x^3}{3x^2+10x-6}\Big)$

c) $\frac{1}{2}$ $(1\times$ l'H$)$ $\Big(\lim\limits_{x\to 0} f(x) = \Big\langle\Big\langle 0\cdot(\pm\infty) \Big\rangle\Big\rangle = \lim\limits_{x\to 0} \frac{x}{\operatorname{tg} 2x} = \Big\langle\Big\langle \frac{0}{0} \Big\rangle\Big\rangle \overset{\text{l'H}}{=} \lim\limits_{x\to 0} \frac{1}{2\frac{1}{\cos^2 2x}}\Big)$

d) 0 $(2\times$ l'H$)$ $\Big(\lim\limits_{x\to 0} f(x) = \Big\langle\Big\langle (\pm\infty)-(\pm\infty) \Big\rangle\Big\rangle = \lim\limits_{x\to 0} \frac{x-\sin x}{x\cdot\sin x} = \Big\langle\Big\langle \frac{0}{0} \Big\rangle\Big\rangle \overset{\text{l'H}}{=} \lim\limits_{x\to 0} \frac{1-\cos x}{\sin x + x\cos x} =$

$= \Big\langle\Big\langle \frac{0}{0} \Big\rangle\Big\rangle \overset{\text{l'H}}{=} \lim\limits_{x\to 0} \frac{\sin x}{\cos x+\cos x - x\sin x}\Big)$

e) $\mathrm{e}^{\frac{3}{2}}$ $(1\times$ l'H$)$ $\Big(x^{3/(x^2-1)} = \exp\Big((3/(x^2-1))\ln x\Big) \overset{\text{ozn.}}{=} \exp(h(x));\ \lim\limits_{x\to 1} h(x) = \lim\limits_{x\to 1} \frac{3\ln x}{x^2-1} =$

$= \Big\langle\Big\langle \frac{0}{0} \Big\rangle\Big\rangle \overset{\text{l'H}}{=} \lim\limits_{x\to 1} \frac{\frac{3}{x}}{2x} = \frac{3}{2}\Big)$

f) $\frac{12}{5}\cdot\frac{\ln 4-\ln 3}{\ln 5-\ln 2}$ $(1\times$ l'H$)$ $\Big(\lim\limits_{x\to 1} f(x) = \Big\langle\Big\langle \frac{0}{0} \Big\rangle\Big\rangle \overset{\text{l'H}}{=} \lim\limits_{x\to 1} \frac{3\cdot 4^x\cdot\ln 4 - 4\cdot 3^x\cdot\ln 3}{5^x\cdot\ln 5 - 5\cdot 2^{x-1}\cdot\ln 2}\Big)$

D/16) a) ∞ $(3\times$ l'H$)$

b) ∞ $(3\times$ l'H$)$

c) 0 $(3\times$ l'H$)$ $\Big(\sqrt[3]{x}\cdot\ln^3 x = \frac{\ln^3 x}{x^{-\frac{1}{3}}}\Big)$

d) 0 $(3\times$ l'H$)$ $\Big(\sqrt[3]{\mathrm{e}^x}\cdot x^3 = \frac{x^3}{\mathrm{e}^{-\frac{x}{3}}}\Big)$

D/17) l'H lze (rozumně) použít a pomůže v těchto případech:

D/1) a) (2× l'H)

b) (2×, příp. 3× l'H)

c) (2× l'H)

d) (5× l'H) (možné, ale zbytečně pracné)

D/3) a) $x_0 = 5$ $(1\times$ l'H$)$;
$x_0 = -\infty$ $(2\times$ l'H$)$

b) $x_0 = 1$ $(2\times$ l'H$)$

c) $x_0 = 1$ $(1\times$ l'H$)$;
$x_0 = -\infty$ $(2\times$ l'H$)$ (oboje po převodu na jeden zlomek)

D/4) a) $x_0 = 1$ $(1\times$ l'H$)$;
$x_0 = +\infty$ $(1\times$ l'H$)$

D/7) b) (1×, příp. 2× l'H)

D/8) a) $x_0 = \frac{\pi}{4}$ $(1\times$ l'H$)$

b) $x_0 = \frac{\pi}{2}$ $(1\times$ l'H$)$ $\Big(\lim\limits_{x\to\frac{\pi}{2}} f(x) = \Big\langle\Big\langle (\mp\infty)-(\mp\infty) \Big\rangle\Big\rangle = \lim\limits_{x\to\frac{\pi}{2}} \frac{\sin x-1}{\cos x} = \Big\langle\Big\langle \frac{0}{0} \Big\rangle\Big\rangle \overset{\text{l'H}}{=}$
$\overset{\text{l'H}}{=} \lim\limits_{x\to\frac{\pi}{2}} \frac{\cos x}{-\sin x} = \frac{0}{-1} = 0\Big)$

c) $x_0 = 0$ $(1\times$ l'H$)$ $\Big(\lim\limits_{x\to 0} f(x) = \Big\langle\Big\langle \frac{0}{0} \Big\rangle\Big\rangle \overset{\text{l'H}}{=} \lim\limits_{x\to 0} \frac{\cos x}{\frac{1}{2}\frac{1}{\sqrt{1+\operatorname{tg} x}}\cdot\frac{1}{\cos^2 x} - \frac{1}{2}\frac{1}{\sqrt{1-\operatorname{tg} x}}\cdot\frac{-1}{\cos^2 x}} =$

$= \lim\limits_{x\to 0} \frac{2\cos^3 x}{\frac{1}{\sqrt{1+\operatorname{tg} x}} + \frac{1}{\sqrt{1-\operatorname{tg} x}}} = \frac{2}{2} = 1\Big)$

D/9) a) $x_0 = -1$ $(1\times$ l'H$)$

b) $x_0 = 0$ $(2\times$ l'H$)$ (možné, ale zbytečně pracné) $\Big(\lim\limits_{x\to 0} f(x) = \Big\langle\Big\langle \frac{0}{0} \Big\rangle\Big\rangle \overset{\text{l'H}}{=} \lim\limits_{x\to 0} \frac{2x}{2\operatorname{tg}\frac{x}{2}\frac{1}{\cos^2\frac{x}{2}}\cdot\frac{1}{2}} =$

$= \lim\limits_{x\to 0} \frac{2x\cos^2\frac{x}{2}}{\operatorname{tg}\frac{x}{2}} = \Big\langle\Big\langle \frac{0}{0} \Big\rangle\Big\rangle \overset{\text{l'H}}{=} \lim\limits_{x\to 0} \frac{2\cos^2\frac{x}{2} + 2x\cdot 2\cos\frac{x}{2}\big(-\sin\frac{x}{2}\big)\cdot\frac{1}{2}}{\frac{1}{\cos^2\frac{x}{2}}\cdot\frac{1}{2}} = \frac{2+0}{\frac{1}{2}} = 4\Big)$

c) $x_0 = 0 \quad (1\times\ \text{l'H}) \qquad \Big(\lim\limits_{x\to 0}\ f(x) = \Big\langle\!\Big\langle 0\cdot(\pm\infty) \Big\rangle\!\Big\rangle = \lim\limits_{x\to 0}\ \dfrac{\sin 3x}{\operatorname{tg} 5x} = \Big\langle\!\Big\langle \dfrac{0}{0} \Big\rangle\!\Big\rangle \overset{\text{l'H}}{=} \lim\limits_{x\to 0}\ \dfrac{3\cos 3x}{5\frac{1}{\cos^2 5x}} = \dfrac{3}{5}\ \Big)$

D/10) a) $x_0 = \pi \quad (1\times\ \text{l'H}) \qquad \Big(f(x) = \mathrm{e}^{h(x)},\ \text{kde } h(x) = \operatorname{cotg} x\cdot\ln(1+3\operatorname{tg} x),\quad \lim\limits_{x\to\pi}\ h(x) = \Big\langle\!\Big\langle \pm\infty\cdot 0 \Big\rangle\!\Big\rangle =$

$= \lim\limits_{x\to\pi}\ \dfrac{\ln(1+3\operatorname{tg} x)}{\operatorname{tg} x} = \Big\langle\!\Big\langle \dfrac{0}{0} \Big\rangle\!\Big\rangle \overset{\text{l'H}}{=} \lim\limits_{x\to\pi}\ \dfrac{\frac{1}{1+3\operatorname{tg} x}\cdot\frac{3}{\cos^2 x}}{\frac{1}{\cos^2 x}} = 3,\ \text{ tedy } \lim\limits_{x\to\pi}\ f(x) = \mathrm{e}^3\ \Big)$

b) $x_0 = 0 \quad (1\times\ \text{l'H}) \qquad \Big(f(x) = \mathrm{e}^{h(x)},\ \text{kde } h(x) = \dfrac{x+1}{x}\cdot\ln(1-2x),\quad \lim\limits_{x\to 0}\ h(x) = \Big\langle\!\Big\langle \pm\infty\cdot 0 \Big\rangle\!\Big\rangle =$

$= \lim\limits_{x\to 0}\ \dfrac{\ln(1-2x)}{\frac{x}{x+1}} = \Big\langle\!\Big\langle \dfrac{0}{0} \Big\rangle\!\Big\rangle \overset{\text{l'H}}{=} \lim\limits_{x\to 0}\ \dfrac{\frac{1}{1-2x}\cdot(-2)}{\frac{x+1-x}{(x+1)^2}} = -2\,,\ \text{ tedy } \lim\limits_{x\to 0}\ f(x) = \mathrm{e}^{-2}$ (při výpočtu limity exponentu h by bylo vhodnější použít l'H jen na limitu podílu $\frac{\ln(1-2x)}{x}$ a získanou limitu vynásobit limitou výrazu $x+1$) $\Big)$

D/11) a) $x_0 = -5 \quad (1\times\ \text{l'H}) \qquad \Big(\lim\limits_{x\to -5}\ f(x) = \Big\langle\!\Big\langle \dfrac{0}{0} \Big\rangle\!\Big\rangle \overset{\text{l'H}}{=} \lim\limits_{x\to -5}\ \dfrac{\frac{1}{2}\,\frac{1}{\sqrt{x^2-9}}\cdot 2x}{1} = -\dfrac{5}{4}\Big)$